TECHNOLOGY STACK

Project Name: EduGenie

Project Type: AI-Powered Learning Assistant

1. INTRODUCTION

EduGenie is an AI-powered learning assistant developed to help students understand and revise educational content. The project uses Google Gemini Generative AI to process user-provided content and generate useful learning responses.

The technology stack consists of programming languages, AI services, development tools and supporting technologies required to build and run the project.

2. PROGRAMMING LANGUAGE

Python

Python is used as the primary programming language for the EduGenie application.

Reasons for using Python:

- Simple and easy-to-understand syntax.
- Suitable for AI and Generative AI applications.
- Provides libraries for API integration.
- Easy to maintain and extend.
- Large developer community and documentation.

3. ARTIFICIAL INTELLIGENCE

Google Gemini Generative AI

Google Gemini is the main AI technology used in EduGenie.

Gemini is responsible for:

- Processing educational content.
- Understanding user requests.
- Generating summaries.
- Providing explanations.
- Generating learning-related responses.

4. AI API INTEGRATION

Google Gemini API

The Gemini API allows the EduGenie application to communicate with the Gemini Generative AI model.

Basic flow:

User Input

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EduGenie Application

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Gemini API

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Google Gemini Model

↓

Generated Response

↓

EduGenie

↓

Student

5. FRONTEND / USER INTERFACE

The user interface provides a simple way for students to interact with EduGenie.

The interface is responsible for:

• Accepting user input.

• Providing a submit/action option.

• Displaying the generated AI response.

• Presenting information in a readable format.

The exact frontend technology depends on the current implementation of the EduGenie project.

6. BACKEND

Python-based application logic is used to manage the core processing of EduGenie.

The backend handles:

- User input.
- Prompt preparation.
- Communication with the Gemini API.
- Receiving AI responses.
- Returning the generated output to the user.

7. DEVELOPMENT TOOLS

The project can be developed and maintained using:

- Visual Studio Code.
- Python development environment.
- Git.
- GitHub.
- Command Line / Terminal.

8. VERSION CONTROL

Git and GitHub

Git is used for tracking changes in the project source code.

GitHub can be used for:

- Storing the project repository.
- Version control.
- Collaboration.
- Project submission.
- Managing project documentation.

9. DEPENDENCY MANAGEMENT

Python package management is used to install and maintain the libraries required by EduGenie.

The project's requirements.txt file can contain the required Python dependencies.

This makes it easier to install the project requirements in another development environment.

10. INTERNET CONNECTIVITY

Internet connectivity is required for communication between EduGenie and the Google Gemini API.

The connection allows the application to send requests to the AI service and receive generated responses.

11. TECHNOLOGY STACK SUMMARY

Layer	Technology
Programming Language	Python
AI Technology	Google Gemini Generative AI
AI Integration	Gemini API
Backend	Python
Frontend	Project-specific user interface
Version Control	Git
Repository	GitHub
Development Tool	Visual Studio Code
Package Management	Python requirements.txt
Connectivity	Internet / API connection

12. CONCLUSION

The EduGenie technology stack combines Python with Google Gemini Generative AI to create an AI-powered learning assistant. Python provides the application logic and API integration, while Google Gemini provides the Generative AI capabilities required to process educational content and generate useful learning responses.